

Coding & Solution

Date	02 OCTOBER 2026
Team ID	6ab7f7bfd083b0e2ee72b13b
Project Name	LegalEase: AI-Powered Legal Document Generator
Maximum Marks	5 Marks

Coding & Solution

This section evaluates the quality and completeness of your implemented code and the overall solution. Your submission should demonstrate working functionality, clean code practices, and alignment with your defined requirements.

Instructions:

1. Ensure your code is well-structured, commented, and follows best practices for your chosen technology stack.
2. The solution should address the core problem statement and implement the key functional requirements.
3. Include all source files, configuration files, and setup instructions needed to run the application.
4. Attach or link to your code repository (e.g., GitHub) in the table below.

Solution Summary

Field	Details
Repository Link / URL	https://github.com/Bavasrikumaran-Gcesalem/LegalEase-AI-Powered-Legal-Document-generator.git
Programming Language(s)	PYTHON
Framework(s) Used	FastAPI, Streamlit, Pydantic

Key Features Implemented	AI-powered legal document generation; predefined and custom document types; structured input for parties, terms and effective date; document preview; document editing; placeholder highlighting; TXT, DOCX and PDF export; custom logo/branding; validation and error handling; privacy and security controls.
Pending / Incomplete Features	No major core functionality is pending. Future enhancements can include additional legal document templates, multilingual document generation, user accounts/document history and further deployment scaling.
Setup / Run Instructions	Clone the GitHub repository, create a Python virtual environment, install dependencies from requirements.txt, configure GEMINI_API_KEY in .env, start the FastAPI backend using uvicorn backend.main:app --reload --port 8000, then start the Streamlit frontend using streamlit run frontend/app.py. For deployment, the project includes Docker, render.yaml, Procfile and Render startup configuration.

Code Quality Checklist

S.No	Criteria	Status (Yes / No)
1	Code is modular and organized into functions / classes	Yes
2	Meaningful variable and function names are used	Yes
3	Code includes comments / documentation where necessary	Yes
4	Error handling is implemented for critical operations	Yes
5	The application runs without critical errors	Yes
6	Code is committed to a version control repository	Yes

Additional Notes / Comments:

LegalEase is implemented as a modular AI-powered legal document generation application. The backend is developed using FastAPI and Pydantic, while the user interface is implemented using Streamlit. The AI document generation module uses Google Gemini through the official Google Gen AI SDK.

The project separates AI generation, document parsing, preview, branding and export functionality into dedicated modules. Document exports are supported in TXT, DOCX and PDF formats. Input validation and structured error handling are implemented for invalid requests, AI failures, invalid logos and document export failures.

The project includes automated tests using pytest and FastAPI TestClient. Source code, configuration files, Docker configuration and Render deployment configuration are maintained in GitHub. The application has also been deployed on Render and is accessible through the provided live deployment URL.